

ExitLight Health Checker Overview

Description

The product I have developed monitors the “health” of emergency exit lighting see, figure 1. The emergency light module I developed can be retrofitted to the existing fixture. It has the ability to automatically switch the fitting off via GPRS using a mobile SIM. When this event occurs a sensor measures the light intensity and transmits this information to AWS cloud server. Depending on your jurisdiction, this should be done periodically, in Ireland this should occur 3 times per year for 30 minutes off time and the final quarter for 3 hours. The battery of the fitting should produce enough energy to illuminate the area for this period. The device also has the capability of incorporating additional sensors i.e. Air Quality sensors, motion sensors, smoke detection sensors, temp, humidity, pressure etc.

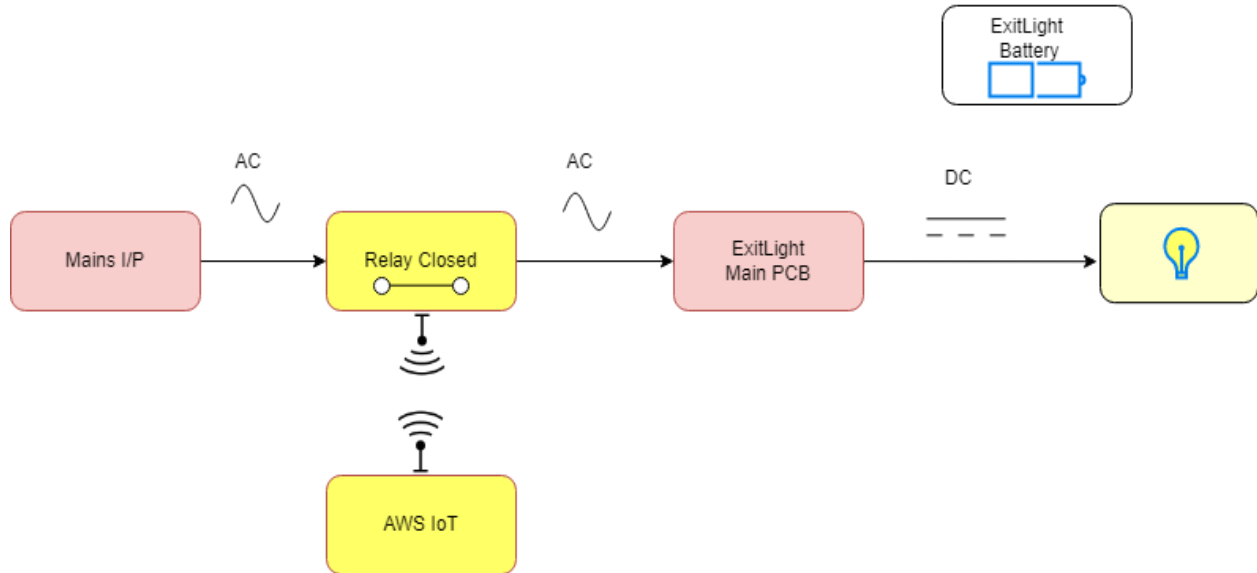


figure1. emergency light fitting

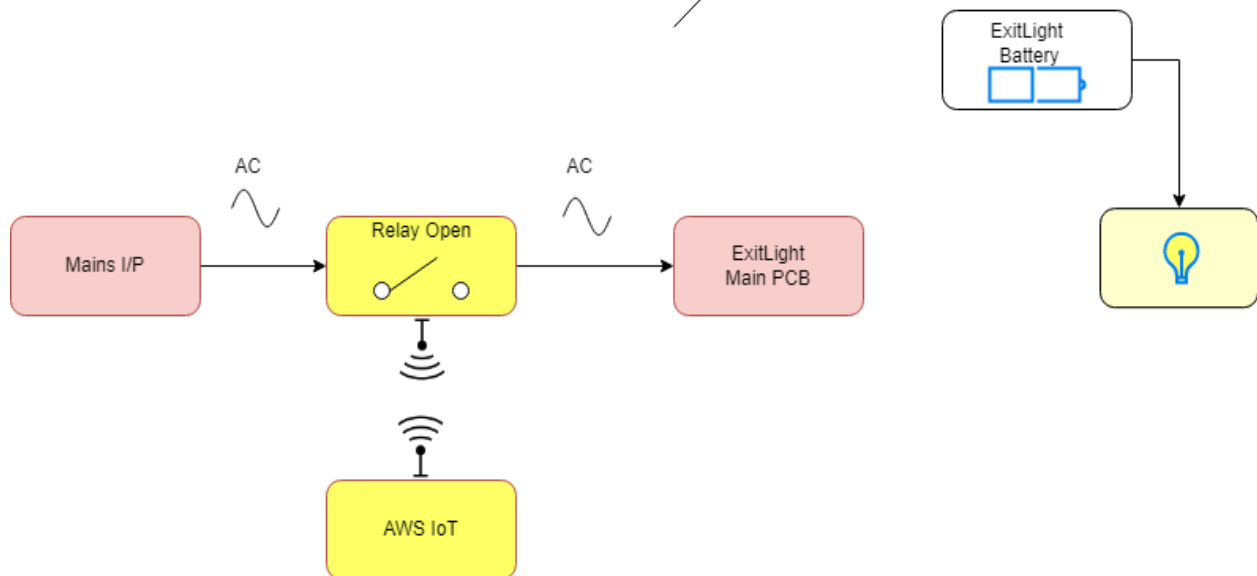
Key Components:

1. MicroController
 - Espressif ESP32 WROOM Controller
2. Communication Interfaces
 - I2C and 2x UART
3. Other components
 - AC/DC converter and regulators
 - Relay SPDT
 - SIM800L Cellular Modem
 - BH1750 Light Sensor
 - External I2C/GPIO interface connector
 - UART programming interface connector
 - Status LED and Power on Indicator
 - Reset button
 - Flash Boot button

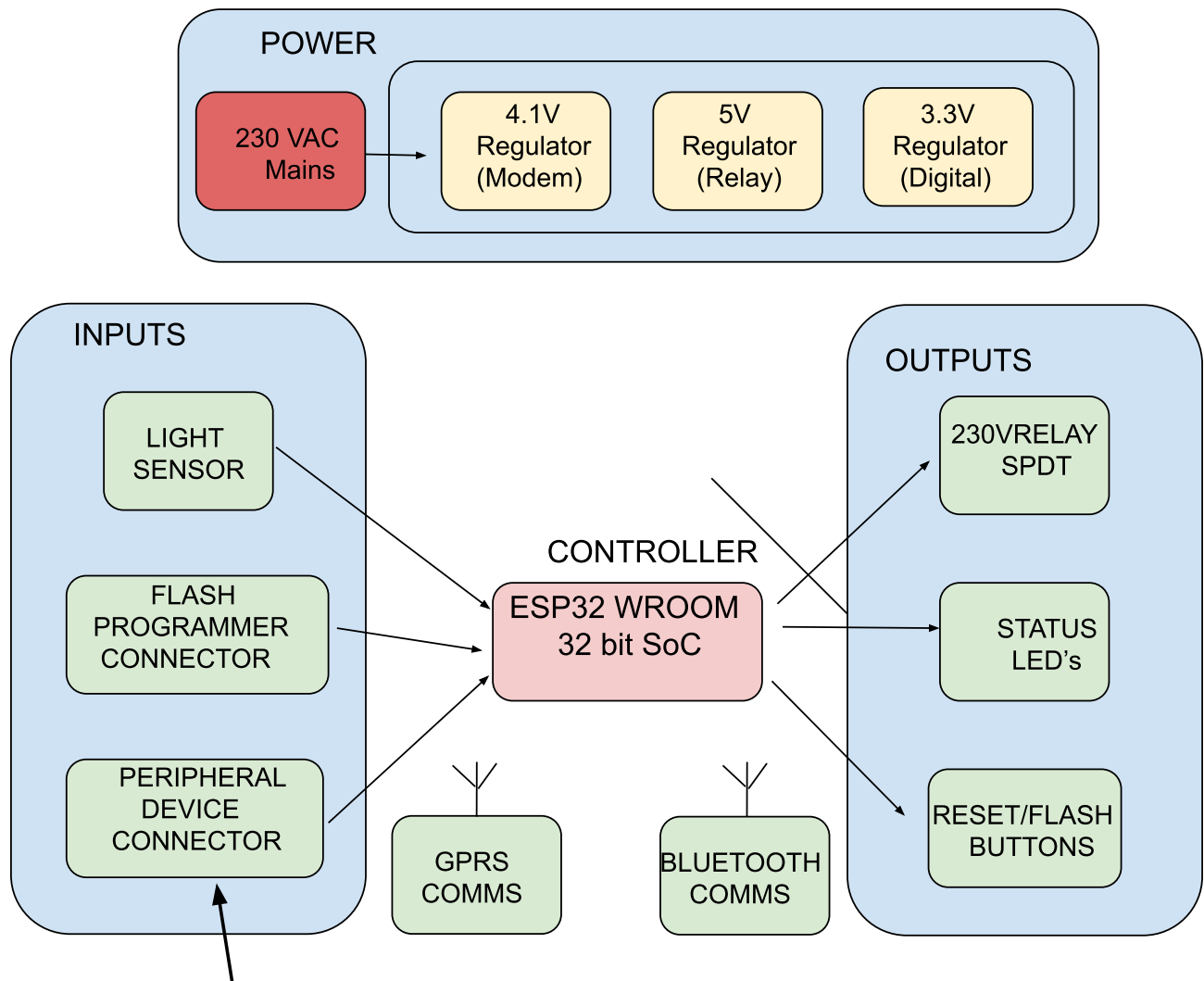
Exit Light Normal Operation



Exit Light Test Mode Operation



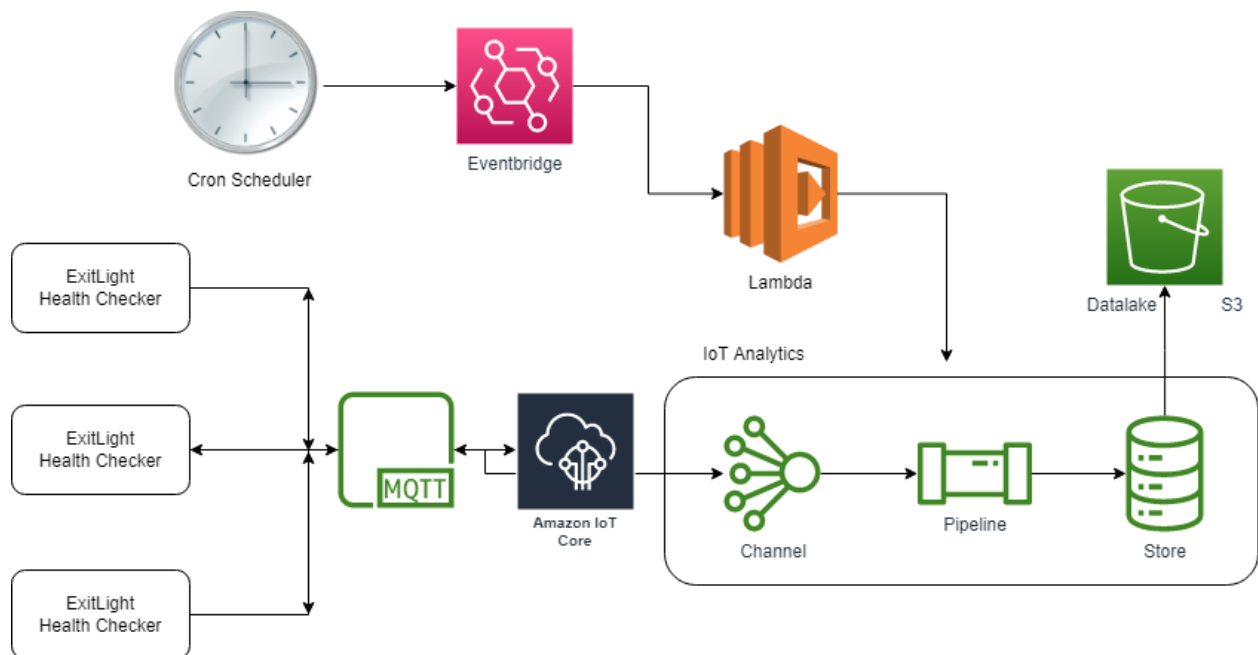
Block Diagram



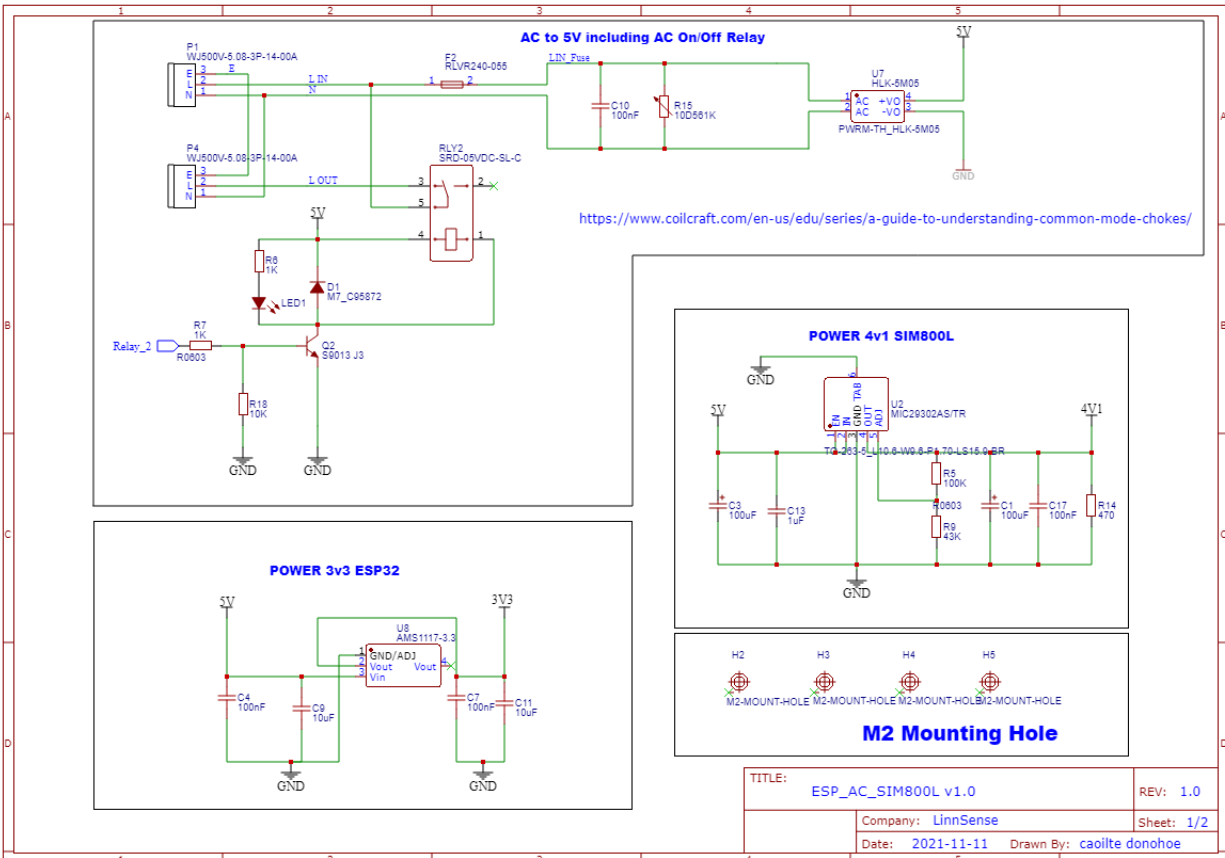
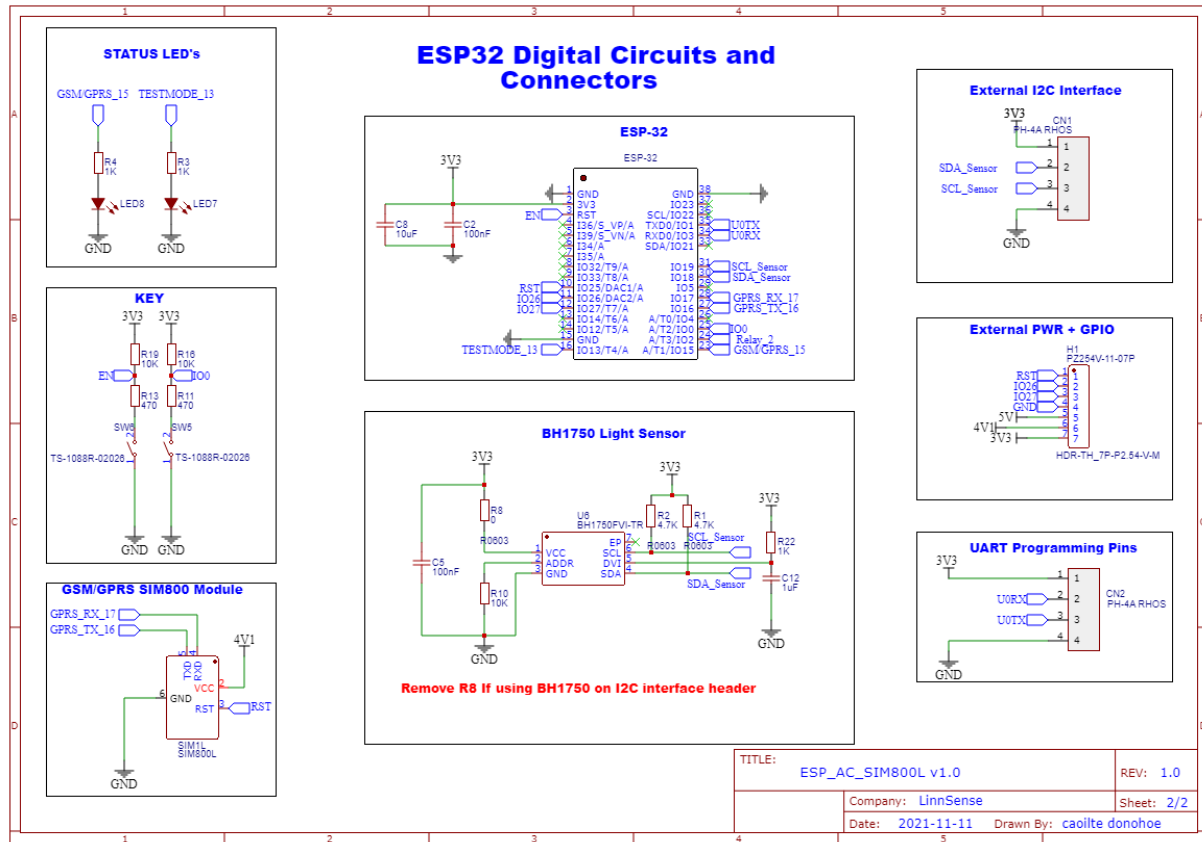
Note: The external peripheral connector can be used to incorporate additional sensors i.e. Air Quality sensors, motion sensors, smoke detection sensors, temp, humidity, pressure etc.

AWS Cloud Services

Flowchart



Schematic Diagram



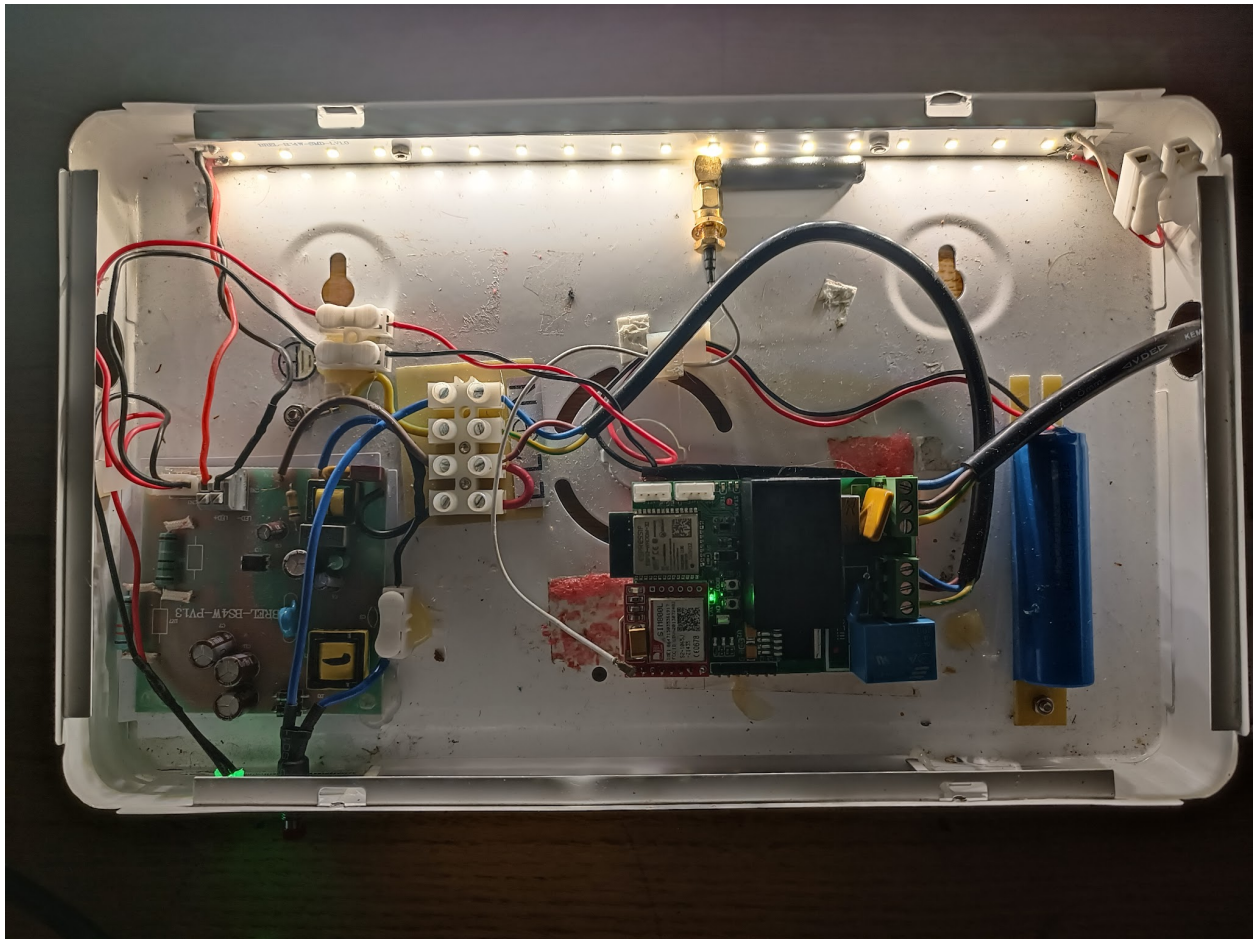
Project Images:



ExitLight Health Checker



OEM ExitLight



OEM ExitLight with Health Checker Installed